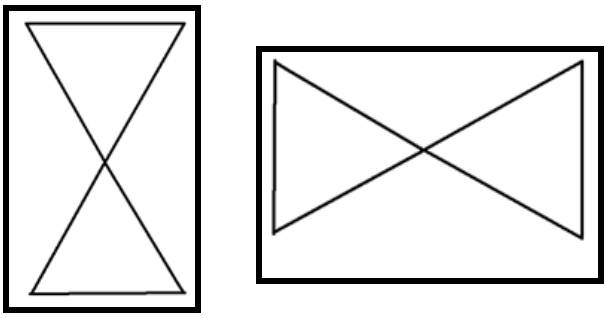
PF X
Project 1 Animations

Problem 1 - FAN PRINTING (Using straight line equation)

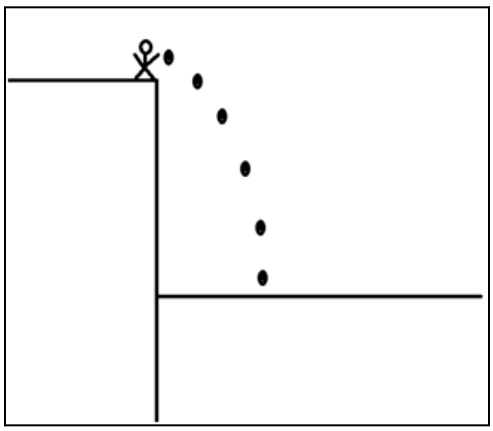
Print the Fan made of the following two shapes printed alternatively, Your Fan should start slowly, slowly and then its speed should rise just like you have switched on the button of the fan.

Here are the shapes: If you print them alternatively it will make it a Rotating Fan.



Problem 2 - STONE THROWING

1. Make this animation: There is a cartoon character standing over a cliff and throwing a stone.
   Hint: http://cnx.org/content/m13837/latest/
   Use the X-equation and Y-equation with respect to time.



2. Make this program so that now the cartoon character is throwing lots of stones (let's say 5) and everyone has associated with itself a resistance, such that now everyone will not fall simultaneously, and everyone has a different horizontal velocity so they will fall with different patterns.

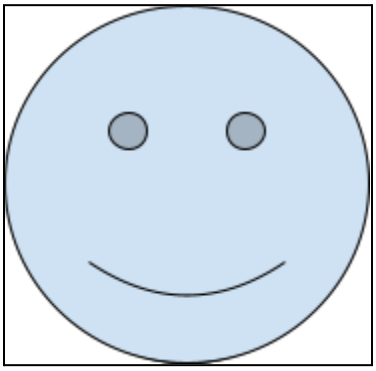
Problem 3 - CIRCLE PRINTING

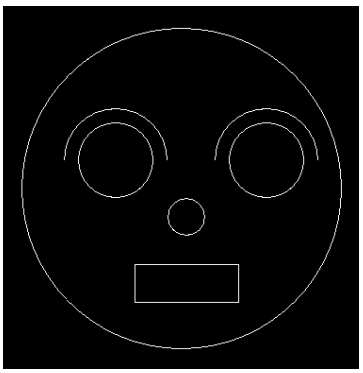
1. Write a program which will first **print the circle with a certain radius R in the middle of the screen** using the circle's equation $x^2+y^2 = R^2$ and then compute the x and y using their parametric equation with respect to Theta. $x = r \cos\theta$ and $y = r \sin\theta$.
   With center at (Cx, Cy) the equation changes to $x = r\cos\theta + Cx$ and $y = r\sin\theta + Cy$.

2. Enhance the first part and **print the circle and then erase it and then Change the Radius** to such that it seems that the circle has been expanded, and then more and more until it reaches to the full-screen. After that it should start shrinking and this process keeps on repeating.

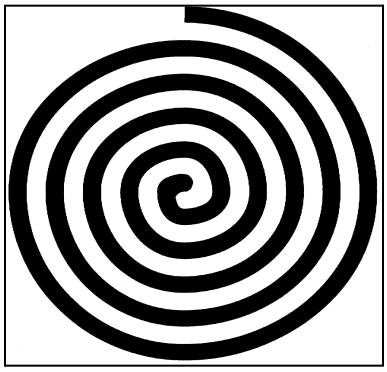
3. Bonus:Make a Balloon and when the user keeps on pressing F(air-filling) make it expand more and more or if the user enters "I" inflate make its radius small and small. And as soon as it is expanded so much that it hits the end outline(It should burst), like all the particles(stars) of the Circle should move towards boundaries, and after hitting they should freely fall downward (to the floor).

4. Print these **SMILEY FACE**



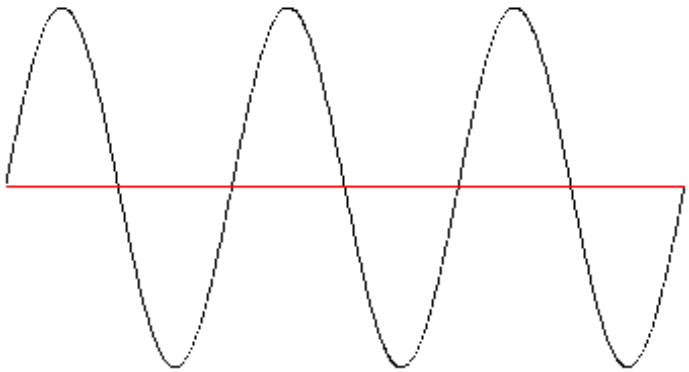


Problem 4 - SPIRAL PRINTING using Polar Equation



The Spiral will not be the exact Spiral as our screen resolution is small.

Problem 5 - SINE WAVE PRINTING



1. Make a function **PrintSineWave(int Cr, int Amp, int W, char sym, int Rows, int Cols)** to display a Sine wave on the screen. It takes the number of waves to print **W**, the x-axis to print around **Cr** and amplitude of the waves **Amp**. Print the Sine wave with fill character or *.

2. Include the starting angle for the sine wave in the above function, and use it to print a Sine wave that looks like it is moving on the screen.
    a. Print moving sine wave with fill character
    b. Print moving sine wave with the letters of your name.

Problem 6 - NAME PRINTING

1. You must have some Name (Lets say Hassan), Your TASK is to print your name on the Screen Character By Character. **Write a function for each Character of your Name**. For Example If your Name is ALI (then you have to make three functions

    Print_A(int centerR, int centerC, int Scale, char Sym)
    Print_L(int centerR, int centerC, int Scale, char Sym)
    Print_I(int Scale))

    which will print the Specific Character depending upon the value of **Scale-value made** up of **pChar(symbol)**.

2. Write a program which will **print your name in a growable fashion** - first it will print a smaller version of your First letter into the middle of the screen then it will slowly slowly grow until it reaches the whole screen dimension. And then it will print the 2nd Character and then the third character.

3. Now extend Part 1, **print your name moving in all directions of the screen such** that the first character will appear vertically up the face of the dimension and start dropping down slowly. And then letter 1 again will appear from Down to Top, then left to right and then right to left. Similarly after this second letter will repeat the same with all 4 directions.

4. BONUS: **All three letters will appear from any three directions and move in random motion.**

read about rand() function http://www.cplusplus.com/reference/cstdlib/rand/

CHALLENGE: CLOCK

Create this clock. The numbers can be printed in the same way we printed characters. The needles are simply lines. The real logic lies in the movement of the needles. You will need to use logical reasoning to convert seconds into degrees of rotation and then translate those angles into coordinate calculations.

BONUS - Spiral Rectangle

I. First it will print by using a symbol (as we did in the class) and then in the same order remove that rectangular Spiral.

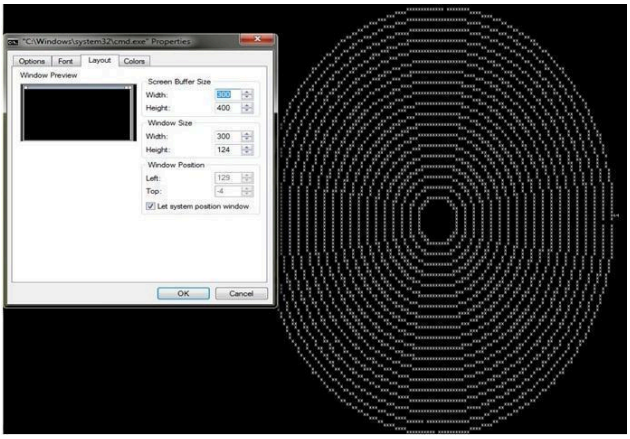
II. Remember the loop containing 4 L2R, T2B, R2L and B2T instructions(functions calls). In that for loop of rectangle printing add the following instruction and see wonders :)

```
for(int rec=1; rec<=NOfRectangles; rec++)
{
        L2R(.......)
        ___________________
        ___________________
        B2T(........)

        Symbol++;
}
```

III. Now your task is to enhance that program to such that it will first print the rectangles and once it reached to the end, it will start removing those rectangles in the backward direction.

IV. Make it generic with SPACING in between the arms of the rectangular spiral.

Note: Think carefully this is not a hard task at all, Just think carefully you will be able to see what you are asked to figure it out.

NOTE : By the way you can change your Resolution: Just like this. Just Click on the left edge of the Console Screen and goto Properties and Change the resolutions (Window Size is the important attribute to change, Screen Buffer Size automatically updates and adjusts itself accordingly).